

observations can be censored either by the death of one individual or if they simply revoke their contract with the insurance company. Insurers can thus observe a multitude of censoring scenarios in which one or more variables can be censored or where multiple factors can cause censoring.

Additionally, a second complexity often encountered in insurance data is the correlation between different – potentially censored – variables of interest to the insurers. This dependence can take the form, for example, of temporal dependence between the occurrence of a specific event and a claim’s development delay as in Zhou and Zhao 2010, Shi, Feng, and Boucher 2016 and Shi and Yang 2018. E. Frees and E. Valdez 2008, Frees, Shi, and Valdez 2009 or Yang and Shi 2019 also illustrate hierarchical dependence within multi-features claims. We can even find cross-sectional dependence between losses and expenses, as in E.W. Frees and E. A. Valdez 1998, or between a claim’s lifetime and its final amount, as in Lopez 2019.

To account for this, many authors in the actuarial literature have used the framework of copulas, i.e., multivariate distribution functions whose marginals are uniformly distributed on the unit interval. Copulas allow a representation of the joint distribution function of multivariate data by separating their marginal distributions from the inherent dependence structure. Examples of their application abound in the statistical literature, and since their introduction by Sklar 1959, many authors have extensively studied and analyzed their properties.

Among the different classes of copulas, Schweizer and Sklar 1983 introduced the Archimedean family, widely applied today in many fields beyond insurance. Contrarily to other copula families, Archimedean copulas are not derived from Sklar’s theorem, as highlighted in Größer and Okhrin 2021. They are characterized instead by a function called the *generator*, $\psi(\cdot) : [0, \infty] \rightarrow [0, 1]$ with $\psi(0) = 1$ and $\lim_{v \rightarrow \infty} \psi(v) = 0$, defined with one or more so-called dependence parameters. The d -variate Archimedean copula $C(\cdot)$ then takes the following form:

$$C(u_1, \dots, u_d) = \psi(\psi^{-1}(u_1) + \dots + \psi^{-1}(u_d)).$$

Given a copula $C(\cdot)$, we can retrieve Kendall’s tau using the relation

$$\tau = 4 \int_0^1 \dots \int_0^1 C(u_1, \dots, u_d) dC(u_1, \dots, u_d) - 1.$$

It is clear that finding an estimator for the generator allows us to directly capture both the shape and strength of dependence for a given set of multivariate observations.

For this purpose, Genest and Rivest 1993 propose a non-parametric approach relying on a study of the bivariate probability integral transformation for Archimedean copulas. Their estimator is, however, only applicable to complete data, seldom encountered in insurance. In addition, this estimator has only been used so far to identify the Archimedean copula best fitted to a given dataset but has not been directly used itself in, for example, simulation studies.

The motivation of our paper is thus two-fold. First, considering the large amounts of data available to insurers and the highly diverse censoring schemes that these datasets can contain, we present a graphical selection procedure for Archimedean copulas based on a non-parametric estimator of the generator. The contribution that we bring to this graphical approach is that we extend it to different flexible censoring scenarios for the data, thereby rendering it well suited to, among others, insurance applications. For this purpose, we base ourselves on the method proposed in Genest and Rivest 1993 and modify it, using results from Wang and Wells 2000 and an estimator for the joint distribution proposed by Akritas and Van Keilegom 2003 in survival analysis in order to account for various censoring schemes. To the best of our knowledge, this estimator has never been used in an actuarial context nor in the application of a non-parametric estimator for Archimedean copulas. In addition, it contrasts with the more naive copula modelling approach that usually consists of fitting a certain number of copulas and comparing their performance using information criteria such as the

Akaike Information Criterion (AIC) or Bayesian Information Criterion (BIC). When using such criteria, we are required to fit all the candidate copula models to the data as a first step, including estimating the dependence parameters. When estimating the generator function, no prior fitting of parametric models is required, and the dependence parameters for various candidate models can be directly derived from the generator, even in the presence of censoring, as we will illustrate in this paper.

Second, we go a step further and, instead of finding the most fitted Archimedean copula, we show how to directly perform simulations from the non-parametric generator. This novel approach allows to use the Archimedean family without assigning a specific copula to the data and thereby limiting ourselves to a named and pre-defined model. This method readily applies to the large amounts of data faced by insurers and frees them from the constraint of defining a parametric model that might miss some of the finer complexities of their datasets. In addition, we show how, in a claims reserving context, performing analysis and predictions directly from the non-parametric estimator of the generator function can lead to more accurate reserves calculations and thereby financial gains for the insurers. Without the constraints of a parametric model that, even though it might be well suited to the data, is unlikely to be a perfect fit, we decrease the risk of prediction inaccuracies arising from model misspecification.

We structure our paper as follows: first, we provide a brief literature review in Section 2. In Section 3, we describe the statistical model for the non-parametric estimator using methods borrowed from survival theory. Section 4 describes the graphical comparison procedure and presents three model validation approaches applicable to different censoring scenarios. In Section 5, we show how to directly generate correlated observations starting from the non-parametric estimator, without formulating any parametric model assumptions. We present an application to a claims reserving framework and a comparison between the graphical and simulation approaches in Section 6, using a Canadian automobile dataset in which policyholders benefit from multiple insurance coverages. This dataset provides a good illustration of a more complex censoring scheme. We conclude our work in Section 7.

2. Literature review

In recent years, a lot of attention has been paid to Archimedean copulas, particularly in finance and insurance. Allowing for very flexible dependence structures, this class of copulas has become popular thanks to their simple form and statistical tractability, among other interesting properties. Notably, many of them allow high-dimensional dependence modeling using only one parameter governing the strength of dependence. Although initially more widely used in life insurance where they arise from frailty models, Archimedean copulas have grown popular in non-life insurance as well.

Thanks to the large quantities of data more easily available, benefits can be reaped from using a non-parametric estimator such as the one proposed by Genest and Rivest 1993 to directly retrieve the generator of Archimedean copula models. We discuss two interesting examples of the application of this estimator in the case of censored data. First, the work of Denuit, Purcaru, and Van Keilegom 2006 combines an estimator for the generator from Wang and Wells 2000 to an estimator of the joint distribution proposed by Akritas 1994 to model the dependence between the allocated loss adjustment expenses and right-censored losses. Their methodology is, however, only applicable to cases where only one variable is subject to censoring, thereby limiting possible replications to more flexible scenarios.

To the best of our knowledge, the only non-parametric approach for more flexible censoring scenarios is the work proposed in Gribkova and Lopez 2015. In their paper, the authors propose a discrete estimator and two smooth estimators for Archimedean copulas and any type of copula applicable to various censoring scenarios. Their discrete estimator extends the empirical copula estimator proposed in Deheuvels 1997 by using random weights in the empirical joint distribution

to account for the bias caused by censored observations. For the two smooth copula estimators, Gribkova and Lopez 2015 introduce kernel functions in addition to the weights of the empirical joint distribution estimator.

In the present paper, we use an estimator situated between those of Denuit, Purcaru, and Van Keilegom 2006 and Gribkova and Lopez 2015. As in Denuit, Purcaru, and Van Keilegom 2006, we focus only on Archimedean copulas and use a non-parametric estimator specifically for Archimedean generators. We, however, seek to allow for the same level of flexibility in the choice of censoring scenarios as Gribkova and Lopez 2015. As such, we extend the work of Denuit, Purcaru, and Van Keilegom 2006 to the case where any or all variables can be subject to censoring by replacing the estimator for the joint distribution with that proposed in Akritas and Van Keilegom 2003.

We obtain a non-parametric estimator for Archimedean copula generators that we then propose to use in two different approaches. First, in a graphical comparison similar to the one used in Denuit, Purcaru, and Van Keilegom 2006 to select the most appropriate parametric copula model for the data. Second, in a simulation routine to generate correlated observations with the same strength and shape of dependence as the original data.

In the first approach, since the choice of the copula model is driven solely by a graphical comparison, we want to use robust approaches to validate it. Goodness-of-fit tests for copulas have been discussed by some authors in the literature, notably Genest, Rémillard, and Beaudoin 2009 who present a review of available tests or Größer and Okhrin 2021 who account for all the latest developments in copula theory, including estimation and testing methods.

However, all these tests only apply to complete data. The literature on goodness-of-fit tests for copulas with censored data is much scarcer. For Archimedean copulas, Lakhal-Chaieb 2010 proposes a test based on a Cramer-von-Mises type distance and Kendall distribution. Wang 2010 presents a test based on a multiple-imputation procedure of the censored data, and Emura, Lin, and Wang 2010 use a cross-ratio function. Some authors have also proposed tests for other families of copulas, such as the likelihood and pseudo-likelihood tests of Yilmaz and Lawless 2011. More recently, Zhou 2021 and Sun, Cheng, and Ding 2023 presented general tests based on information matrices applicable to various copula families. In addition to their three non-parametric copula estimators, Gribkova and Lopez 2015 also propose a test similar to that of Lakhal-Chaieb 2010 that can be used with different kinds of distances such as Kolmogorov-Smirnov, Cramer-von-Mises or the square-root of a quadratic integrated distance.

In this paper, we use three approaches to validate the non-parametric estimation of the Archimedean copula generator. More specifically, we use an omnibus procedure based on the likelihood function, a method based on the L^2 -norm distance, inspired by Lakhal-Chaieb 2010 and Gribkova and Lopez 2015, and the test proposed by Wang 2010.

Our second approach, namely direct simulations from the non-parametric estimator of the generator function, is inspired by the work of Hofert 2008 who described various sampling schemes for nested Archimedean copulas. In the various algorithms presented in his work, the author however assumes a known Archimedean copula family with a well-defined parametric generator, or generators for which the Laplace-Stieltjes transforms are known. Ridout 2009 proposes an algorithm that can generate random observations from a distribution specified by its inverse Laplace-Stieltjes transform. In this work, we adapt Ridout 2009's algorithm to the case where the Laplace-Stieltjes transform is defined by a non-parametric function, then use this combined with some algorithms presented by Hofert 2008 to sample from our original data, without specifying a parametric copula model.

3. Statistical Model

In this section, we describe an approach to retrieve the Archimedean copula generator $\psi(\cdot)$ in the case of flexible censoring. More specifically, we present a non-parametric estimator building on the work of Akritas and Van Keilegom 2003 in survival analysis. The advantage of using such a

non-parametric estimator lies in the possibility of directly tailoring the model to the data, even in the presence of complex censoring.

We begin by defining the notation before describing the estimator. For clarity, we focus solely on the bivariate case, but the notation and model described hereafter can easily be extended to a multivariate framework.

3.1 Notation

We define the following model components:

- $\mathbf{T} = (T_1, T_2)$ with $T_i \in \mathbb{R}^+$ for $i = 1, 2$ is the initial vector of variables of interest. They can represent times-to-event, losses, etc. To avoid identifiability issues with the copula, we assume these variables to be continuous;
- $\mathbf{X} = (X_1, X_2)$ with $X_i \in \mathbb{R}^+$ for $i = 1, 2$ is the vector of censoring variables, such that $\mathbf{T}$ and $\mathbf{X}$ are independent;
- $\boldsymbol{\omega} = (\omega_1, \omega_2)$, with $\omega_i > 0$ for $i = 1, 2$ are constants acting as additional censoring factors on $\mathbf{T}$. We call these the *limits*.
- $\mathbf{Y} = (Y_1, Y_2)$ where $Y_i = \min(T_i, X_i, \omega_i)$ for $i = 1, 2$ is the observed bivariate vector, taking censoring of $\mathbf{T}$ into account;
- $\boldsymbol{\Delta} = (\mathbb{1}_{[Y_1=T_1]}, \mathbb{1}_{[Y_2=T_2]})$ is the vector of censoring indicators.

To illustrate the difference between the censoring variable X and the limit ω , consider a study of the time to recovery after administration of a treatment of patients suffering from a medical condition. The time to recovery, denoted by T , is censored if the patient dies, with the time to death being the censoring variable X . In addition, health professionals might consider the treatment ineffective if the patient does not recover after a given amount of time, even if he is still alive. Further treatments will then need to be considered. In this example, the time elapsed until the treatment is deemed to have failed is the limit ω . Even if the patient is still alive, i.e., $T > X$, the imposition of ω will censor his time to recovery. The observed delay $Y = \min(T, X, \omega)$ is of interest to medical practitioners. Y represents the delay until the first event, being either full recovery, death, or need for additional treatments.

Further applications can be found in finance, insurance, and other fields. Section 6 focuses on an application to bivariate claims delays.

3.2 Non-parametric estimator of the generator

We now describe the non-parametric approach leading to the identification of the copula linking the pairs of censored variables (Y_1, Y_2) .

Assuming flexible censoring scenarios where either one or both variables can be subject to random censoring, we follow the methodology laid out in Akritas and Van Keilegom 2003 and work with the following extension of Beran 1981's estimators for the conditional distributions of Y_1 (resp. Y_2) given $Y_2 = y_2$ (resp. $Y_1 = y_1$):

$$\hat{F}_{1|2}(y_1|y_2) = 1 - \prod_{Y_{i1} \leq y_1, \Delta_{i1}=1} \left(1 - \frac{W_{ni2}(y_2; h_n)}{\sum_{j=1}^n W_{nj2}(y_2; h_n) \mathbb{1}_{Y_{j1} \geq Y_{i1}}} \right), \quad (1)$$

where y_2 must be uncensored and

$$W_{ni}(y; h_n) = \begin{cases} \frac{k\left(\frac{y-Y_{i2}}{h_n}\right)}{\sum_{\Delta_{j1}=1} k\left(\frac{y-Y_{j2}}{h_n}\right)}, & \text{if } \Delta_{i2} = 1 \\ 0, & \text{if } \Delta_{i2} = 0. \end{cases}$$

Here, $k(\cdot)$ is a known kernel function and $\{h_n\}$ is the bandwidth: a sequence of positive constants such that $h_n \rightarrow 0$ as $n \rightarrow \infty$. We work with a similar estimator for $P[Y_2 \leq y_2 | Y_1 = y_1] = \hat{F}_{2|1}(y_2 | y_1)$.

The estimator of the joint distribution proposed in this context by Akritas and Van Keilegom 2003 is then given by:

$$\hat{F}(\mathbf{y}) = w(\mathbf{y}) \int_0^{y_2} \hat{F}_{1|2}(y_1 | z_2) d\tilde{F}_2(z_2) + (1 - w(\mathbf{y})) \int_0^{y_1} \hat{F}_{2|1}(y_2 | z_1) d\tilde{F}_1(z_1), \quad (2)$$

where $\tilde{F}_1$ and $\tilde{F}_2$ are the marginal estimators of Kaplan and Meier (1958):

$$\tilde{F}_j(y_i) = 1 - \prod_{Y_{ij} \leq y_i, \Delta_{ij}=1} \left(1 - \frac{1}{n - i + 1}\right), \quad j = 1, 2.$$

The weights $w(\mathbf{y})$ minimise the mean-squared error of $\hat{F}(\mathbf{y})$. Equation (2) is the estimator that, combined with Genest and Rivest 1993's approach, allows us to retrieve the generator of the copulas in the presence of different censoring scenarios. We start from Genest and Rivest 1993' proposition laid out just below.

Proposition 3.1 (Genest and Rivest 1993). *Let $S_1(t_1)$ and $S_2(t_2)$ be marginal survival functions whose dependence function $C(\cdot)$ is of the form*

$$C(S_1(t_1), S_2(t_2)) = \psi \left\{ \psi^{-1}(S_1(t_1)) + \psi^{-1}(S_2(t_2)) \right\},$$

with $\psi(\cdot) : [0, \infty] \rightarrow [0, 1]$ with $\psi(0) = 1$ and $\lim_{v \rightarrow \infty} \psi(v) = 0$. Let

$$U = \frac{\psi^{-1}(S_1(t_1))}{\psi^{-1}(S_1(t_1)) + \psi^{-1}(S_2(t_2))} \quad \text{and} \quad V = \psi \left\{ \psi^{-1}(S_1(t_1)) + \psi^{-1}(S_2(t_2)) \right\} = C(S_1(t_1), S_2(t_2)).$$

Then,

1. U is uniformly distributed on $(0, 1)$;
2. V follows the Kendall distribution defined as $K(v) = v - \frac{\psi^{-1}(v)}{[\psi^{-1}]^{(1)}(v)}$, for $0 < v \leq 1$; and
3. U and V are independent.

Genest and Rivest 1993 suggest the following estimator for the Kendall distribution in the presence of complete data:

$$\hat{K}_n(v) = \frac{1}{n} \# \{i | v_i \leq v\},$$

where $\#$ denotes the cardinality of a set.

We cannot use this estimator in the presence of censored data. Instead, we choose to apply the extension proposed by Wang and Wells 2000 to estimate the Kendall distribution by

$$\hat{K}_n(v) = \int_0^\infty \int_0^\infty \mathbb{1}_{[\hat{F}(\mathbf{y}) \leq v]} \hat{F}(d\mathbf{y}), \quad (3)$$

in which we insert (2) as the estimator of the joint distribution, allowing for all possible censoring scenarios.

Finally, we insert $\hat{K}_n(v)$ in the estimator for the Archimedean generator proposed by Genest and Rivest 1993:

$$\hat{\psi}_n^{-1}(v) = \exp \left\{ \int_{v_0}^v \frac{1}{t - \hat{K}_n(t)} dt \right\}, \quad (4)$$

with $0 < v_0 < 1$ an arbitrarily chosen constant. Thanks to the use of Equation (2), we can retrieve the generator non-parametrically, regardless of the level of censoring present in the data.

In Section 4 and 5, we now present two different methods to use $\hat{\psi}_n(v)$ to perform further analysis and predictions.

4. Graphical comparison

Given the empirical estimator $\hat{K}_n(v)$, we can select the parametric copula model best fitted to the data at hand using the graphical procedure introduced by Genest and Rivest 1993. Let the univariate function $\lambda(v)$ be defined from the Kendall's distribution as

$$\lambda(v) = v - K(v) = \frac{\psi^{-1}(v)}{[\psi^{-1}]^{(1)}(v)}.$$

An estimator of $\lambda(v)$ is easily retrieved by

$$\hat{\lambda}_n(v) = v - \hat{K}_n(v). \quad (5)$$

The idea is now to graphically compare the empirical estimator $\hat{\lambda}_n(v)$ to the corresponding $\lambda_{\hat{\alpha}}(v)$ of different competing copula models under consideration, where $\hat{\alpha}$ is the dependence parameter of the copula. Knowing $\hat{\lambda}_n(v)$, we proceed as described below:

1. Estimate Kendall's tau using the following equality proposed by Genest and Rivest 1993:

$$\hat{\tau} = 4 \int_0^1 \hat{\lambda}_n(v) dv = 3 - 4 \int_0^1 \hat{K}_n(v) dv \quad (6)$$

2. Given the relationship between Kendall's tau and the dependence parameter α for some Archimedean copulas, retrieve these copula parameters using

$$\hat{\alpha} = g^{-1}(\hat{\tau}). \quad (7)$$

Table 11 in Appendix 1 provides the functions $g(\cdot)$ as well as other specific information for some popular Archimedean copulas.

3. Given $\hat{\alpha}$, trace the curves of $\lambda_{\hat{\alpha}}(v)$ for some competing copula models, using again the information provided in Appendix 1.
4. Select the parametric copula model m for which the curve of $\lambda_{\hat{\alpha}_m}(v)$ is closest to that of $\hat{\lambda}_n(v)$.

One important benefit of this graphical procedure is that, contrarily to the more classical model selection using information criteria, it only requires to estimate $\hat{\lambda}_n(v)$. From this estimator, we can directly evaluate on the graph what type of Archimedean copula might best fit our data by observing the shape of the $\hat{\lambda}_n(v)$ curve. We also easily retrieve the copula parameter using Equation (7). There is no need to fit different copula models beforehand and compare them using AIC and BIC. Estimating $\hat{\lambda}_n(v)$ is enough to identify the most appropriate model, even in the presence of incomplete data.

4.1 Results validation

While practical and easy to implement, the estimator described in Section 3.2 is non-parametric, and the choice of the copula model is driven at this stage by a graphical analysis only. Goodness-of-fit tests for copulas are rare in the literature, even more so in the presence of censoring. To validate our graphical choice and increase the level of confidence in our selected model, we briefly describe three approaches in this section. More details on each of them can be found in [Appendix 2](#).

A first way to verify whether the selected copula model is appropriate for the data at hand is to compare the dependence parameter $\hat{\alpha}$ estimated from Equation (7) to its equivalent $\hat{\alpha}^*$, obtained from an optimisation procedure. This is the omnibus procedure, or maximum pseudo-likelihood procedure that consists in a semi-parametric optimization that substitutes empirical versions of the marginal distributions in the (parametric) likelihood function of the copula model, $L(\cdot)$. This procedure has been used, for example, in Genest, Ghoudi, and Rivest 1995 and Denuit, Purcaru, and Van Keilegom 2006.

The second approach is based on the L^2 -norm. Following the original idea of Wang and Wells 2000, we verify that the distance between the non-parametric estimator of the Kendall distribution $\hat{K}_n(v)$ and $K_{\hat{\alpha}_m}(v)$ for the selected model m is the smallest in comparison with the distances to the Kendall distributions of all other candidate models. We propose the following parametric bootstrap procedure, inspired by Lakhal-Chaieb 2010 and Gribkova and Lopez 2015 to select the most appropriate copula model based on a pseudo p -value:

Step 1. Compute $\hat{K}_n(v)$ for the data at hand and, for the M candidate models under consideration, estimate $\hat{\alpha}_m$, for $m = 1, \dots, M$.

Step 2. For each candidate model $C_{\hat{\alpha}_m}$, generate B censored samples of size n .

Step 3. For each of the B samples, estimate $\hat{\alpha}_{m,b}$ and $D(\hat{\alpha}_{m,b})$.

Step 4. For each model m , obtain the pseudo p -value as

$$p_m = \frac{1}{B} \sum_{b=1}^B \mathbb{1}[\min_l D(\hat{\alpha}_{b,l}) > D(\hat{\alpha}_{b,m})]$$

with $m \neq l$. The copula model m best fitted to the data is the one for which the pseudo p -value is the smallest. In this case, we can interpret p_m as the number of times among B replications that copula model m is selected as most fitted for the data at hand.

The third and final approach to validate the chosen copula model is a more robust statistical test proposed by Wang 2010. This goodness-of-fit test bases its premise on Proposition 3.1 from Genest and Rivest 1993: under the null hypothesis that an Archimedean copula can model the dependence between the pairs of observations (T_1, T_2) with generator $\psi(\cdot)$, the correlation coefficient between the variables U and V is null. Wang 2010 describes the test in details and extends it to censored data using a multiple imputation procedure.

4.2 Simulation study

Using simulation studies, we now illustrate the graphical approach described in this section. We provide a few examples to showcase the graphical model selection procedure and the validation approaches discussed in Section 4.1.

Independence case. The first way to validate the results from the estimator for the generator in Section 3 is to analyze how it handles independent data. Setting Kendall's tau equal to 0, each of the four copulas should converge to the independence copula. This is the case if $\alpha = 0$ for both the Clayton and Frank copulas and $\alpha = 1$ for the Gumbel and Joe copulas.

We simulate samples of size $n = 1000$ from each of the four copulas with $\tau = 0$. We then use the methodology laid out in Section 3 to estimate $\hat{K}_n(\mathbf{v})$, $\hat{\tau}$ and, finally, $\hat{\alpha}$ for each copula. Figure 1 presents the graphical comparison between the average of the 1000 non-parametric curves $\hat{\lambda}_n(\mathbf{v})$ and the corresponding $\lambda_{\hat{\alpha}(\mathbf{v})}$ for four candidate models, namely the Clayton, Frank, Gumbel and Joe copulas. We also present the average estimates of $\hat{\alpha}$ over the 1000 simulations in Table 1.

Figure 1 shows that the curves for the four candidate parametric models as well as the non-parametric curve, all converge towards the same curve on the plot, i.e. the curve of the independence copula, and it is hard to differentiate between them. Table 1 confirms these results by showing that the parameter estimates of the Clayton and Frank copula tend to 0, while those of the Gumbel and Joe copulas tend to 1. Our approach using Akritas and Van Keilegom 2003's estimator for the joint distribution provides adequate estimates of the dependence parameters when $\tau = 0$.

Table 1. Average estimates of $\hat{\alpha}$ for 1000 simulations when $\tau = 0$.

	Clayton	Frank	Gumbel	Joe
$\hat{\alpha}$	0.0249	0.0352	1.0163	1.0150

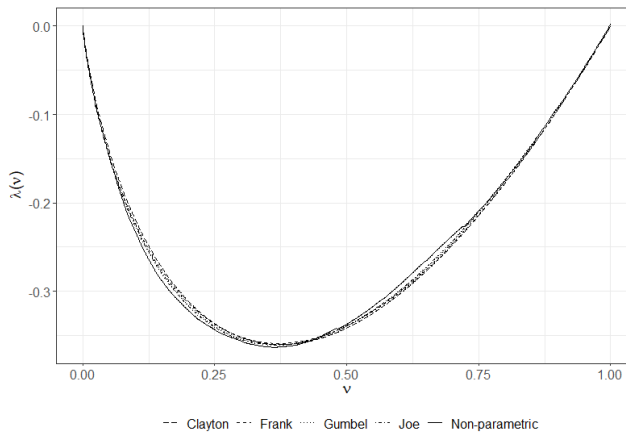


Figure 1. Graphical comparison of the estimated $\hat{\lambda}(\mathbf{v})$ functions for the independent samples.

Graphical comparison. We now move away from the independence scenario and focus on the non-parametric estimator of $\hat{\lambda}_n(\mathbf{v})$ used for the graphical comparison.

We simulate bivariate samples (T_1, T_2) of $n = 1000$ observations from the Clayton, Gumbel, Frank, and Joe copulas. In each case, we select the dependence parameter such that Kendall's tau equals 0.4, and we work with unit-exponential distributions for the marginals. We simulate the vector of censoring variables (X_1, X_2) from exponential distributions with their parameters set such that 20% of observations have at least one censored component. We are in a double-censoring scenario where one or both variables can be censored.

Figure 2 displays the empirical estimator $\hat{\lambda}_n(\mathbf{v})$ and the curves of $\lambda_{\hat{\alpha}}(\mathbf{v})$ for the four copulas under consideration, for each of the four simulated samples. In each plot, the continuous curve depicts the non-parametric estimator presented in Equation (5). For all, $\hat{\lambda}_n(\mathbf{v})$ appears to be closest to the curve of the correct copula, i.e., the copula from which the sample was simulated. These graphical

comparisons suggest that the graphical approach proposed in this section performs quite well under flexible censoring scenarios.

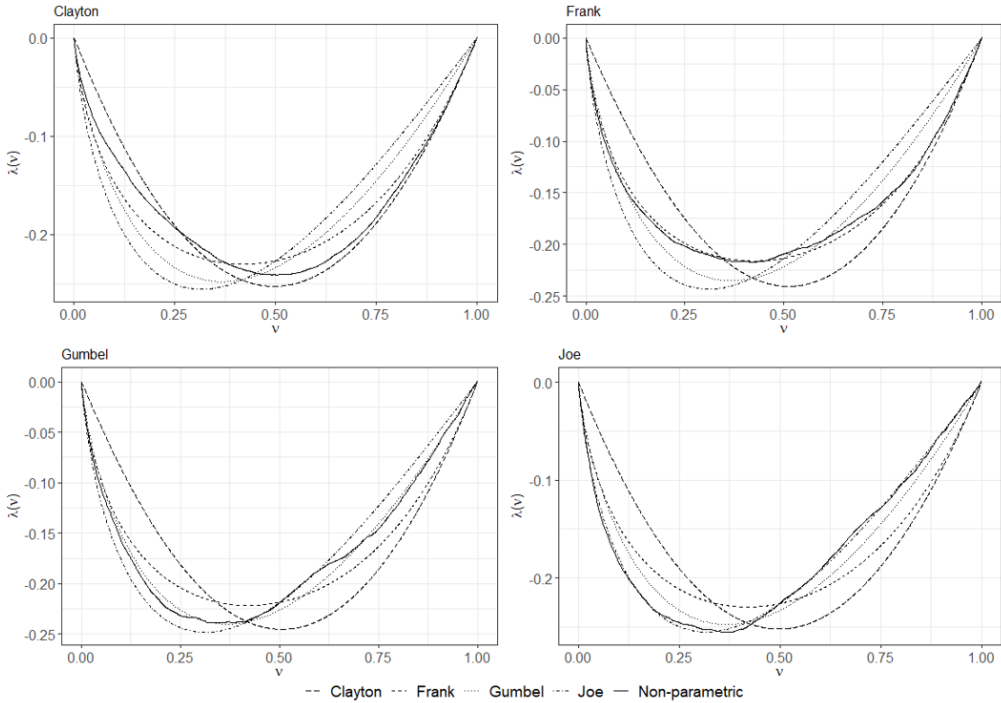


Figure 2. Graphical comparison of the estimated $\hat{\lambda}(v)$ functions for the four simulated samples. We simulate $n = 1000$ bivariate observations from a Clayton (top left), Frank (top right), Gumbel (bottom left) and Joe (bottom right) copulas.

Omnibus procedure. We begin validating the results from Figure 2 with the omnibus procedure. In addition to the double-censoring scenario considered here, we add two other censoring schemes: one with complete data (no censoring) and one where we allow for only one censored variable. We work again for these two additional scenarios with simulated samples of size $n = 1000$, Kendall's tau equal to 0.4, and unit-exponential margins. For the single-censoring scenario, we simulate the censoring variable following an exponential distribution such that around 20% of observations have a censored component, similar to the double-censoring scheme.

Table 2 shows the results of performing the omnibus comparison between $\hat{\alpha}$ and $\hat{\alpha}^*$ one thousand times. Each value in this table represents the percentage of simulations in which the competing models were not considered best-fitted for the sample in terms of the difference between the estimated dependence parameters. We observe, for example, that when we simulate a sample from the Clayton copula with $\tau = 0.4$ in a double-censoring scenario, only 276 simulations out of a thousand reject the Clayton model as best-fitted. Among these 276 simulations, 188 opt for a Joe model, 71 for a Gumbel model, and 17 for a Frank model.

Table 13 in Appendix 3 shows the results of one of these simulations when $\tau = 0.4$ by comparing the dependence parameters $\hat{\alpha}$ estimated with Equation 7 to the corresponding $\hat{\alpha}^*$ for the omnibus procedure.

The results displayed in both Table 13 and Appendix 3 illustrate that the omnibus procedure is a good first indicator of the appropriateness of the copula model selected via the graphical procedure.

Table 2. Percentage of simulations in which different candidate copula models are rejected with the omnibus procedure.

Scenario	True copula	Candidate model			
		Clayton	Frank	Gumbel	Joe
No censoring	Frank	0.876	0.322	0.853	0.949
	Joe	0.897	0.680	0.845	0.578
Single-censoring	Frank	0.880	0.274	0.884	0.962
	Joe	0.892	0.844	0.868	0.396
Double-censoring	Clayton	0.276	0.983	0.929	0.822
	Frank	0.865	0.392	0.947	0.796
	Gumbel	0.906	0.992	0.370	0.732
	Joe	0.936	0.942	0.962	0.160

L^2 -norm. Next, we perform a second study in which we simulate bivariate samples of $n = 500$ observations from various copulas with Kendall's tau equal to 0.2, 0.4, and 0.6. We use the same exponential distributions of parameters equal to one for the marginal distributions of (T_1, T_2) . We again consider three censoring schemes: one with no censoring, one where we allow only one variable to be censored, and one where both variables are subject to censoring. For the double-censoring scheme, we also analyze the independence case.

When working with complete data, we use Genest and Rivest 1993's estimator $\hat{K}_n(\mathbf{v})$ to compute $D(\hat{\alpha})$:

$$\hat{K}_n(\mathbf{v}) = \frac{1}{n} \# \{i | v_i \leq \mathbf{v}\},$$

with

$$v_i = \frac{1}{n-1} \# \{(t_{1,(j)}, t_{2,(j)}) | t_{1,(j)} < t_{1,(i)}, t_{2,(j)} < t_{2,(i)}\}.$$

For the single-censoring scenario, we use a Wang and Wells 2000's estimator $\hat{K}_n(\mathbf{v})$ similar to Section 3.2, but with a different estimator for the joint distribution $\hat{F}(\mathbf{y})$. Namely, we compute the estimator proposed by Akritas 1994 for single-censoring scenarios. Assuming that T_1 is the variable that can be censored by X_1 , we have:

$$\begin{aligned} \hat{F}(y_1, t_2) &= \int_0^{t_2} \hat{F}_{Y_1|T_2}(y_1|z) d\hat{F}_{T_2}(z) \\ &= \frac{1}{n} \sum_{k=1}^n \mathbb{1}[0 \leq t_{2,k} \leq t_2] \hat{F}_{Y_1|T_2}(y_1|t_{2,k}), \end{aligned}$$

where $\hat{F}_{Y_1|T_2}(y_1|z)$ is Beran 1981's estimator from Equation (1).

For the two censoring scenarios, we simulate the censoring variables using exponential distributions with parameters such that at least 20% of observations have one (or two) censored components. For each sample, we then perform the bootstrap procedure described earlier to obtain estimates of the L^2 -norm distance between $\hat{K}_n(\mathbf{v})$ estimated non-parametrically for the data and the corresponding $K_{\hat{\alpha}}(\mathbf{v})$ for different candidate models.

Table 3 presents the results of 1000 simulations for each sample. Each value in this table corresponds to the pseudo p -value from the bootstrap procedure and should be understood as the percentage of bootstrap simulations in which the candidate model was not selected as the best model for the

simulated sample. That is the percentage of simulations for which the L^2 -norm distance between $\hat{K}_n(\nu)$ and $K_{\hat{\alpha}}(\nu)$ for the candidate model was not the smallest one compared to the other models under consideration. We use samples from both Frank and Joe copulas in the no-censoring and single-censoring schemes. At the same time, in the double-censoring scenario, we analyze samples from the Clayton, Frank, Gumbel, and Joe copulas.

We observe that in all three scenarios for each copula, the pseudo p -value is the smallest for the candidate model corresponding to the true copula. For example, in the double-censoring scenario, when we simulate the data from a Frank copula with Kendall's tau equal to 0.4, we observe that the Frank copula is rejected as an appropriate candidate model in just 1.5% of the bootstrapped simulations performed. The Gumbel model is rejected in 98.5% of simulations, and the Clayton and Joe copulas are rejected in all simulations. In addition, we observe that as the value of Kendall's tau increases, the percentage of rejection for the correct model decreases. For the Frank sample in the double-censoring scenario, 8.7% of the bootstrap simulations reject the Frank model when $\tau = 0.2$. This percentage decreases to 1.5% when $\tau = 0.4$ and to 0.9% when $\tau = 0.6$. We observe similar results and trends in all scenarios for each copula considered. This showcases the strength of our test.

In addition, for the doubly-censored scenario, we simulate samples from each of the four copulas with Kendall's tau equal to 0. This is equivalent to simulating four bivariate samples from the independence copula. We observe that the results of the bootstrap simulations are much more random. For the Joe sample, for example, 45% of simulations reject the Clayton model, all reject the Frank model, 80% reject the Gumbel model, and 75% reject the Joe model. This is to be expected since, when $\tau = 0$, all four copulas converge towards the same independence copula. The estimates that we thereby get for $K_{\hat{\alpha}}(\nu)$ and $\lambda_{\hat{\alpha}}(\nu)$ all converge towards the same functions. The choice of the smallest distance between $\hat{K}_n(\nu)$ and $K_{\hat{\alpha}}(\nu)$ for the different candidate models thus becomes much more random. This was already well illustrated in Figure 1 that shows the plot of $\hat{\lambda}(\nu)$ and those of $\lambda_{\hat{\alpha}}(\nu)$ for the four candidate models considered—the curves for the Clayton, Frank, Gumbel, and Joe copula overlap.

Goodness-of-fit test. Using the same censoring scenarios, marginal distributions, and copulas as for the L^2 -norm with Kendall's tau equal to 0.2, 0.4, and 0.6, we now apply Wang 2010's goodness-of-fit test described in Appendix 2. We present the results of 1000 simulations for simulated samples of size $n = 200$ in Table 4. For the single-censoring and double-censoring scenarios, we use $M = 5$ imputed datasets.

We observe similar results to those obtained by Wang 2010 and to those from Table 3. When dependence is weak, the test does not always accurately predict the best model for the data. In particular, we see that in the single-censoring scenario when the sample comes from the Frank copula with $\tau = 0.2$, even though the lowest percentage of rejection of the null hypothesis is correctly observed when the candidate model is the Frank copula, this percentage is also relatively low for the three alternative models. As already observed in Wang 2010, the power of the test, however, increases when we increase the sample size. To illustrate this, we present in Figure 3 what happens to the p -values in the no-censoring scenario when the true model is the Frank copula with $\tau = 0.4$ when we progressively increase the sample size from $n = 200$ to $n = 2000$. We work with increasing steps of 200 observations; in each case, we perform 1000 simulations. The strength of the test increases quite steeply with the sample size. For the Frank candidate model, the resulting p -value starts at a low level of approximately 5%, as was reported in Table 4, and quickly tends to 0 as we increase the sample size. The Gumbel copula displays a low p -value of less than 10% for the smallest sample size. However, it greatly increases such that for a sample of 2000 observations, it is correctly rejected by over 85% of simulations. The Joe and Clayton candidate models display similar increasing trends, even though they started with a higher p -value for small sample sizes than the Gumbel copula.

Similarly to the results of the L^2 -norm in Table 3, we also observe in Table 4 that the strength

Table 3. Pseudo p -values for different candidate copula models under different censoring scenarios.

Scenario	True copula	τ	Candidate model			
			Clayton	Frank	Gumbel	Joe
No censoring	Frank	0.2	1.000	0.091	0.909	1.000
		0.4	1.000	0.067	0.948	0.985
		0.6	1.000	0.001	0.999	1.000
	Joe	0.2	1.000	0.999	0.924	0.077
		0.4	1.000	1.000	0.998	0.002
		0.6	1.000	1.000	0.999	0.001
single-censoring	Frank	0.2	1.000	0.078	0.923	0.999
		0.4	1.000	0.054	0.946	1.000
		0.6	1.000	0.012	0.999	0.989
	Joe	0.2	1.000	1.000	0.928	0.072
		0.4	1.000	1.000	0.987	0.013
		0.6	1.000	1.000	0.999	0.001
double-censoring	Clayton	0.0	0.470	0.890	0.790	0.850
		0.2	0.170	0.831	0.999	1.000
		0.4	0.079	0.921	1.000	1.000
		0.6	0.000	1.000	1.000	1.000
	Frank	0.0	0.230	1.000	0.770	1.000
		0.2	0.986	0.087	0.928	0.999
		0.4	1.000	0.015	0.985	1.000
		0.6	1.000	0.009	0.991	1.000
	Gumbel	0.0	0.620	1.000	1.000	0.038
		0.2	1.000	0.970	0.168	0.862
		0.4	1.000	0.975	0.135	0.890
		0.6	1.000	0.986	0.057	0.957
	Joe	0.0	0.450	1.000	0.800	0.750
		0.2	1.000	1.000	0.843	0.157
		0.4	1.000	1.000	0.929	0.071
		0.6	1.000	1.000	0.949	0.051

of the test increases with the level of dependence. In all scenarios and for all copula models, the percentage of rejection of the null hypothesis increases with τ when the null hypothesis is incorrectly specified and decreases with τ when the true and null copula coincide.

Impact of the limit. In Section 3.1, we introduced a limit ω_i for $i = 1, 2$ acting as an additional censoring layer. Depending on the context, these limits can be imposed by practitioners, authorities, etc. They can be the maximum time before treatment is considered ineffective, even if the patient is still alive, or the maximum payment delay authorized for a claim for example.

In this section, we conduct a simulation study to assess the impact of ω on the strength and form of dependence between two censored variables.

Consider a bivariate vector of time-to-events $\mathbf{T} = (T_1, T_2)$ with log-normal distributions such that $\mu_{T_1} = 8$, $\mu_{T_2} = 7$, $\sigma_{T_1} = 1$ and $\sigma_{T_2} = 3$. We set the correlation ρ between T_1 and T_2 at 0.35. Let $\mathbf{X} = (X_1, X_2)$ be the censoring vector with X_1 independent from X_2 . Both X_1 and X_2 are log-normally distributed, with parameters μ_{X_1} and μ_{X_2} chosen to allow for different levels of censorship. We also set $\sigma_{X_1} = 1$ and $\sigma_{X_2} = 1$. We perform three simulation studies: one with low levels of censorship for both variables (approx. 2% and 1% for, respectively, T_1 and T_2), one with medium levels of censoring (approx. 25% and 40%) and one with higher levels of censorship (approx. 65% and 75% for, respectively, T_1 and T_2). Finally, consider the limits ω_1 and ω_2 such that we observe the bivariate vector $\mathbf{Y} = (Y_1, Y_2) = (\min(T_1, X_1, \omega_1), \min(T_2, X_2, \omega_2))$.

We use the strategy laid out in Section 3 to derive the joint distribution (2) from Beran's estimators.

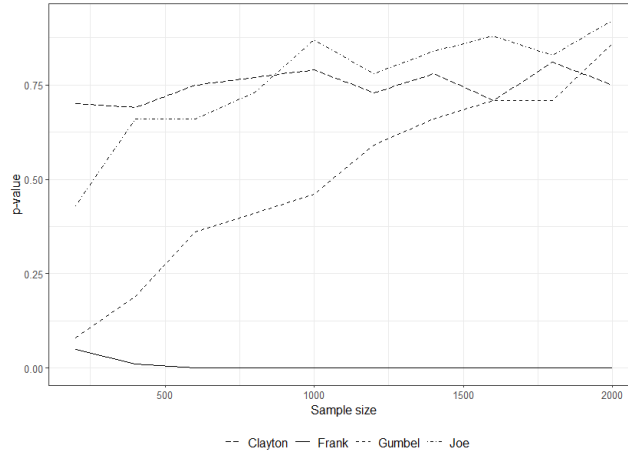


Figure 3. Evolution of the p -values of competing models based on 1000 simulations for varying sample sizes simulated from the Frank copula with Kendall's tau equal to 0.4 in a no censoring scenario.

Table 4. Percentage of rejection of the null hypothesis for different copulas $n = 200$

Scenario	True copula	τ	Copula under H_0			
			Clayton	Frank	Gumbel	Joe
No censoring	Frank	0.2	0.351	0.082	0.030	0.074
		0.4	0.728	0.058	0.086	0.679
		0.6	0.859	0.022	0.196	0.956
	Joe	0.2	0.814	0.206	0.248	0.022
		0.4	0.922	0.776	0.724	0.018
		0.6	0.944	0.970	0.942	0.003
single-censoring	Frank	0.2	0.142	0.032	0.229	0.142
		0.4	0.712	0.084	0.800	1.000
		0.6	0.760	0.000	0.968	0.990
	Joe	0.2	0.720	0.030	0.052	0.016
		0.4	0.970	0.232	0.081	0.007
		0.6	0.980	0.400	0.131	0.004
double-censoring	Clayton	0.2	0.097	0.031	0.512	0.239
		0.4	0.021	0.203	0.835	0.945
		0.6	0.016	0.645	0.968	0.958
	Frank	0.2	0.391	0.022	0.057	0.029
		0.4	0.810	0.012	0.850	0.999
		0.6	0.944	0.001	0.975	1.000
	Gumbel	0.2	0.456	0.082	0.010	0.012
		0.4	0.848	0.172	0.030	0.051
		0.6	0.962	0.458	0.025	0.775
	Joe	0.2	0.990	0.468	0.535	0.081
		0.4	1.000	0.530	0.727	0.002
		0.6	1.000	0.880	0.843	0.000

As in Akritas and Van Keilegom 2003, we choose the weights $w(\gamma) = 0.5$ for the sake of simplicity although better results could be obtained by optimizing them. Using the estimated joint distribution, we then derive Kendall's tau and estimate $\hat{\lambda}_n(v)$ for a sample of $n = 500$ observations. We use different quantiles from the distributions of Y_1 and Y_2 to set the values of the limits ω_1 and ω_2 .

Figure 4 displays the results of decreasing the limit as well as the level of censoring on the form

of dependence via the function $\hat{\lambda}_n(v)$ and the strength of dependence via Kendall's tau. We observe that neither the limit nor the level of censoring seem to have an impact on the form of dependence: in all cases, the shape of the generator function remains similar. As we increase the level of censoring and the limit, we notice that the shape of dependence becomes even more pronounced. This is in line with the results from Tables 3 and 4 where we noted that, even though both tests were able to correctly choose the most appropriate copula for a given sample and low dependence levels, their strength increased with Kendall's tau.

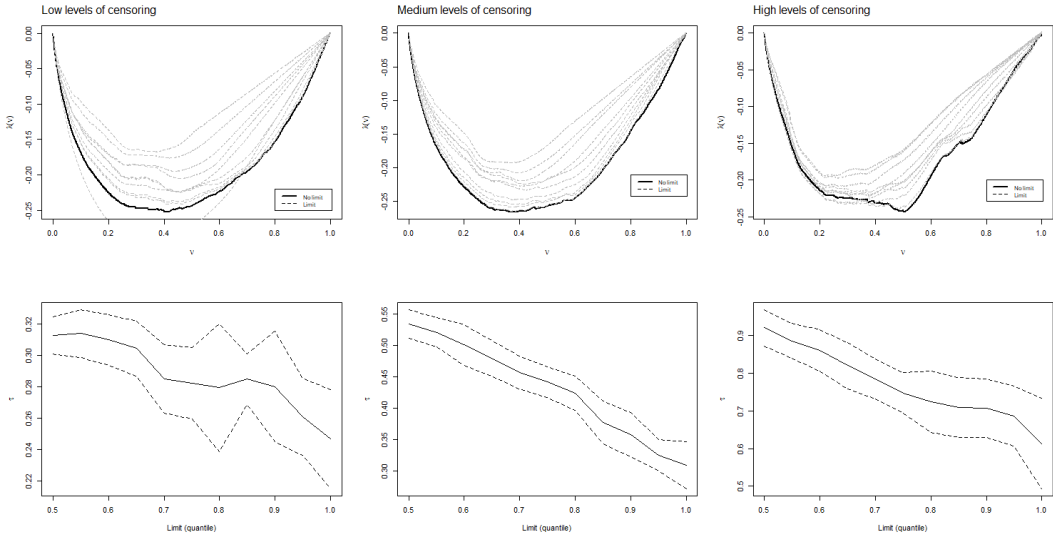


Figure 4. Plots of $\lambda(v)$ and Kendall's tau for the joint distribution using different limits and different levels of censoring. Low, medium and high levels of censoring correspond to, approximately, 5%, 30% and 75%.

5. Simulation from $\hat{\psi}_n(\cdot)$

Although the graphical procedure described in Section 4 can lead to the selection of an adequate parametric copula model for a given dataset, we might be interested in working directly from the non-parametric estimator of the generator $\hat{\psi}_n(v)$ obtained in Section 3. This could be particularly interesting in cases where the graphical comparison does not lead to a clear copula, for example if the curve of $\hat{\lambda}_n(v)$ lies somewhere between the curves of two candidate models.

When this occurs, an attractive alternative would consist in foregoing any parametric model and avoid labelling the copula. Besides the obvious advantage of removing any parametric constraints linked to the already-defined models, working directly from $\hat{\psi}_n(v)$ implies that we do not have to select a pool of candidate models nor perform any goodness-of-fit tests or model validations. Instead, we would directly simulate from the generator to retrieve the starting correlated variables. This is possible when considering Bernstein's theorem:

Theorem 5.1 (Bernstein). *A function $\psi(\cdot)$ is strictly monotonic and $\psi(0) = 1$ if and only if it can be written as*

$$\psi(v) = \mathcal{L}_{\Theta}(v), \quad (8)$$

where $\mathcal{L}(\cdot)$ is the Laplace-Stieltjes transform of the strictly positive random variable Θ .

As highlighted by Hofert 2008, if we are able to sample from the inverse Laplace-Stieltjes transform of the generator, then we can use existing algorithms such as the one proposed in Marshall and Olkin 1988 to simulate d -variate vectors of observations $(U_1, \dots, U_d)$ from the Archimedean copula from which the data originates.

Algorithm 1: Marshall, Olkin

- 1: Generate a random observation θ from the distribution with Laplace transform ψ .
 - 2: For $i = 1, \dots, d$, generate i.i.d. $X_i \sim U(0, 1)$.
 - 3: Return $(U_1, \dots, U_d)$ where $U_i = \psi(-\log(X_i)/\theta)$, for $i = 1, \dots, d$.
-

Ridout 2009 proposes a method for the first step of Algorithm 1, namely the RLAPTRANS Algorithm. This algorithm is a standard modification of the classic Newton-Raphson method to ensure convergence when generating random numbers with an inverse Laplace transform. We slightly modify it in order to be able to use it with our non-parametric estimator function $\hat{\psi}_n(v)$. The main steps are summarized below but more details can be consulted in Ridout 2009.

Algorithm 2: RLAPTRANS

- 1: Generate n independent $U(0, 1)$ observations and sort them: $u_{(1)} < \dots < u_{(n)}$.
 - 2: Find a value θ_{max} to serve as upper bound, i.e. $\psi^{-1}(\theta_{max}) \geq u_{(n)}$.
 - 3: Set the lower bound $\theta_L = \theta_{(i-1)}$ and upper bound $\theta_U = \theta_{max}$. Repeat the modified Newton-Raphson procedure for $i = 1, \dots, n$ to obtain the ordered sample $\theta_{(1)}, \dots, \theta_{(n)}$.
 - 4: Permute the ordered sample randomly to obtain the unordered sample $\theta_1, \dots, \theta_n$.
-

Thanks to this approach, we can relax any assumption related to a parametric model and are able to directly use the data to perform new simulations and, in a next step, predictions. We also note that since any censoring schemes present in the original data is already accounted for in the non-parametric estimator $\hat{\psi}_n(v)$, this method works well regardless of the censoring scenario at hand.

5.1 Simulation study

We now illustrate the use of the non-parametric generator function $\hat{\psi}_n(v)$ instead of a pre-defined parametric copula model by means of some simulation studies.

In a first example, we generate a dataset of $n = 500$ bivariate observations from a Clayton copula with Kendall's tau equal to 0.4, marginal exponential distributions of parameter equal to 1 for (T_1, T_2) and with the censoring variables (X_1, X_2) also following exponential distributions with parameters set such that around 20% of observations have at least one censored component. Figure 5 displays the original non-parametric estimator $\hat{\lambda}_n(v)$ for the Clayton data in black compared to the results of direct simulations from $\hat{\psi}_n(v)$ in blue. In total, we simulated 100 new sets of size $n = 500$, i.e. 50 000 new pairs of observations (Y_1, Y_2) . For each of these new sets, we used again the approach laid out in Section 3 to retrieve the estimator $\lambda_{i, \hat{\psi}_n}(v)$, for $i = 1, \dots, 1000$. The dashed blue curve depicted in Figure 5 is the average of these 1000 curves, i.e. $\bar{\lambda}_{\hat{\psi}_n}(v)$ and the two dotted curves represent the 95% confidence interval of these simulations.

We observe that our non-parametric simulations provide results that are very close to the original data. This is further confirmed by the plots displayed in Figure 6 that show the scatter plots of the original Clayton data in black (top left) and of the average simulations in blue (top right), as well

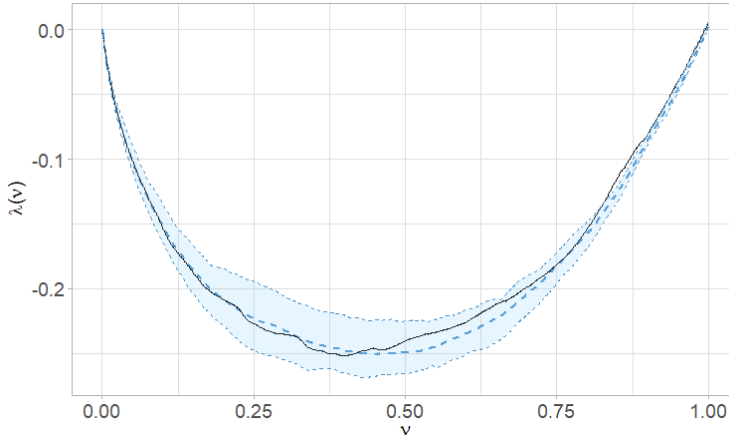


Figure 5. Comparison between $\hat{\lambda}_n(v)$ (black continuous curve) initially obtained for the original data and the average $\bar{\lambda}_{\hat{\Psi}_n}(v)$ (dashed blue curve) obtained across 1000 simulations using Algorithms 2 and 1. The shaded area represents the 95% confidence interval for the simulations.

as a comparison between the original and simulated densities for Y_1 (bottom left) and Y_2 (bottom right). On the density plots, the black continuous curves represent the original densities and the black vertical curves the average observed values for both variables. The dashed blue curve and vertical blue curves show, respectively, the simulated densities and simulated average values.

From the curves in Figure 5 and the scatterplots in Figure 6, we see that the simulated sets have the same shape and strength of dependence as the original data. Both scatterplots display the typical shape expected for data issued from a Clayton copula, with a clear concentration of points on the bottom left corner of the plot. The similar levels of concentration of points observed in both plots, as well as the similar shapes of the curves in Figure 5 further show that the dependence between our simulated bivariate observations is also of similar strength to that of the original couples. In addition, we observe in Figure 6 that the simulated marginal densities for Y_1 and Y_2 in, respectively, the bottom left and bottom right plots, are similar to the original densities shown in black. This further illustrates that directly simulating from the non-parametric generator allows to generate very similar sets of observations, without the need to specify a parametric model nor perform any model validation tests.

In Figure 7, we show similar simulation results when the data originates from a Frank, Gumbel and Joe copulas for different values of Kendall's tau, using heat maps for the simulated scatterplots. In each case, we observe that the sets simulated using the non-parametric generator display the same shapes and strength of dependence as the original data. When $\tau = 0.25$, the heat maps of the simulated data points are rather diffuse and it is harder to recognise the copula from which the data originates. As τ increases, however, the shapes of the heat maps become clearer and look more and more like the typical shapes expected for the different copula models. For the data issued from the Frank copula on the first row of plots, we observe that the points gather more on the diagonal as the level of dependence increases. For the Gumbel copula, we clearly see the higher densities in the bottom left and top right corners of the plots becoming more apparent as τ increases. Finally, for the Joe copula, we see the density becoming more important on the top right corner of the plot as Kendall's tau becomes larger.

These simulations illustrate how using the RLAPTRANS algorithm in combination with an algorithm such as the one proposed by Marshall and Olkin 1988 allows to efficiently simulate correlated pairs of observations from an Archimedean family without the need to define a parametric

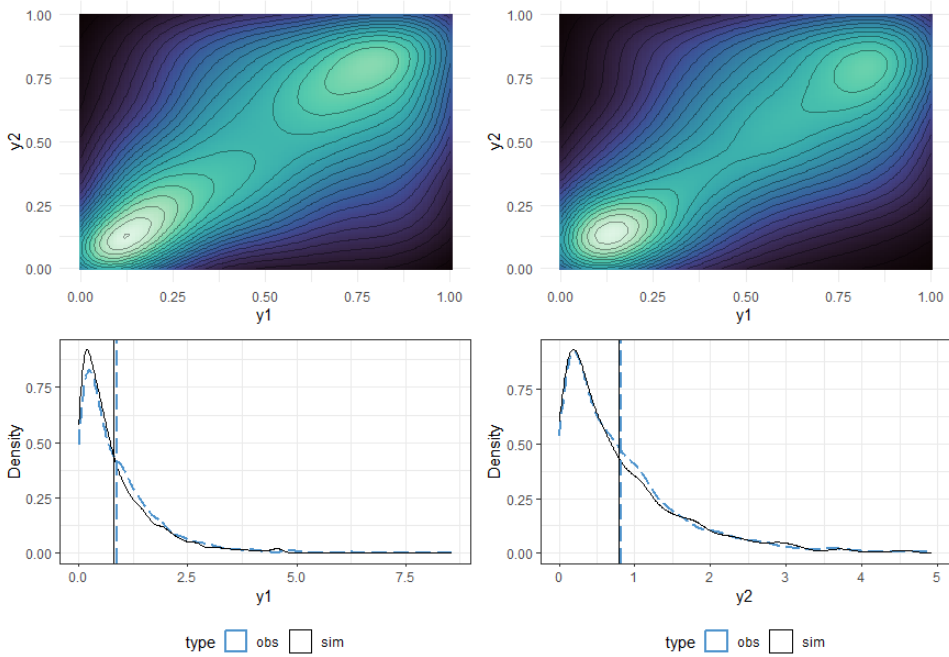


Figure 6. Scatterplots of the original data issued from a Clayton copula (top left) and of the re-simulated sample (top right), and comparison between the original and simulated densities of Y_1 (bottom left) and Y_2 (bottom right). On the density plots, the black continuous curves show the original densities and the black vertical line the observed average values, while the dashed blue curves and vertical blue lines represent, respectively, the simulated densities and simulated average values.

copula model. Using the non-parametric estimator $\hat{\psi}_n(v)$, we are easily able to simulate new pairs of observations with the same shape and strength of dependence as the original sample. These new simulations can then be used in prediction models, as we illustrate in the next Section.

6. Application to automobile insurance claims

In this section, we apply the approach described in Section 3 to claims reserving, using a recent Canadian automobile insurance dataset.

6.1 Data description

The data comprises over 600 000 claims that occurred between January 2015 and June 2021. For each of these claims, we have information related to the policyholder, the vehicle driven or the claim itself. In addition, each policy in force in this dataset provides four different coverages to its holder, and each claim can fall under one or more of these coverages: the Accident Benefits coverage (loss of revenue, funeral expenses for the insured), the Bodily Injury coverage (loss revenue or medical expenses for a third party), the Vehicle Damage coverage (damages incurred to the insured's or a third party's vehicle) and the Loss of Use coverage (temporary replacement vehicle and alternative means of transportation). Table 5 provides some insights into the dynamics of our portfolio by presenting the relative importance of each of these four coverages in terms of the proportion of claims, total cost and total reserve that they represent. We observe that even though the Vehicle Damage and Loss of Use coverages are triggered by much more claims than the Accident Benefits and Bodily Injury coverages, they barely represent together 15% of the total reserve, taking January 1st, 2019 as valuation date. Although less than 6% of all claims trigger the Bodily Injury coverage, this coverage

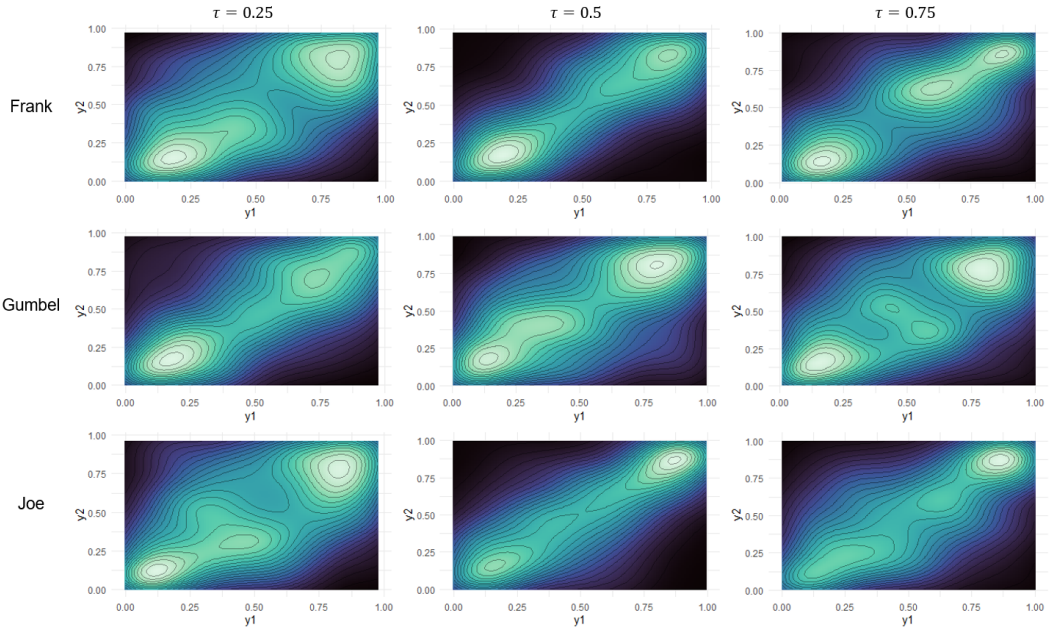


Figure 7. Heat maps for the scatterplots of data simulated from the non-parametric estimator of the generator function, when the original sample comes from either the Frank, Gumbel or Joe copulas for different values of Kendall's tau.

alones represent more than half of the total portfolio reserve.

Table 5. Weight of each coverage in the portfolio. The percentages with respect to the total reserve are calculated by taking the 1st of January 2019 as valuation date

Coverage	% of claims	% of total cost	% of total reserve
Accident Benefits	9.42	12.82	30
Bodily Injury	5.70	13.13	55
Vehicle Damage	96.39	70.44	14
Loss of Use	51.89	3.61	< 1

Table 6 further illustrates the importance of the Accident Benefits and Bodily Injury coverages for the insurer by presenting some descriptive statistics for the severity of payments of all four coverages. Bodily Injury and Accident Benefits claims have the largest average payments and present large values in the higher quantiles, indicating heavy-tailed distributions. In the rest of this section, we thereby choose to focus on these two coverages only, which are clearly the most important cost-wise in the portfolio.

6.2 Notation

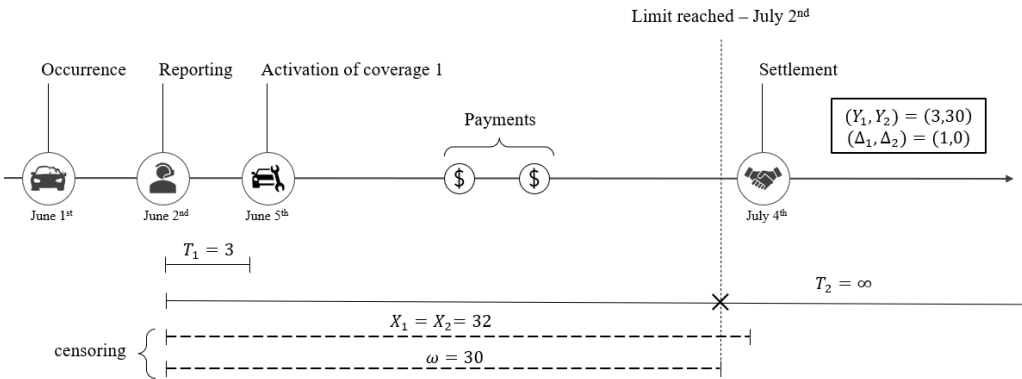
Our goal is to model the dependence between them using their activation delays. These are defined as the time elapsed between the reporting date of the claim and the date at which the insurer first triggers the coverage in the claims management system. This is the date the insurer records that the claim falls under that specific coverage. We denote the vector of activation delays by $\mathbf{T} = (T_1, T_2)$. If the claim settles before one of the coverages is triggered, then the delay for this coverage is censored. The censoring variable, that we denote by $\mathbf{X} = (X_1, X_2)$, represents the claim settlement delay, i.e. the delay between the reporting date and settlement date of the claim. In addition, policyholders may

Table 6. Descriptive statistics for the severity of payments of the four insurance coverages

Coverage	Mean	Std. dev.	Quantiles				Max.
			0.5	0.75	0.95	0.99	
Accident Benefits	12,386	53,561	3,215	6,909	47,757	127,896	2,435,334
Bodily Injury	23,271	76,027	4,000	15,150	98,449	322,612	2,039,570
Vehicle Damage	5,040	8,121	2,605	5,830	17,984	40,611	149,399
Loss of Use	545	620	419	714	1,000	2,336	52,777

be unable to receive compensation for a specific coverage if a certain amount of time has passed since reporting. This limit on the delays can depend on the nature of the claims, local or governmental regulations, or even on company-level rules of the insurer. Both internal and external agents can set it. This is our limit ω_i , for $i = 1, 2$ introduced in Section 3.1. Suppose a coverage has not been triggered yet when this limit passes. In that case, its activation delay becomes censored, even if the claim remains open.

Figure 8 illustrates the typical development of a claim and the notation introduced above. In this example, a claim occurred on June 1st and was reported the following day. On June 5th, the insurer records that the claim has impacted coverage 1. Recording the delays in days, the activation delay for this coverage is then $T_1 = 3$. Assuming a limit of $\omega_1 = \omega_2 = 30$ days on the activation delays, coverage 2 becomes censored on July 2nd, even if the claim is still open. It settles two days later, leading to $X_1 = X_2 = 32$. The observed delays are then given by $(Y_1, Y_2) = (\min(T_1, X_1, \omega_1), \min(T_2, X_2, \omega_2)) = (\min(3, 32, 30), \min(\infty, 32, 30)) = (3, 30)$.

**Figure 8.** Typical claim development and illustration of the notation used.

Considering the observed activation delays shown (in periods of six months) in Table 7 for the Accident Benefits and Bodily Injury coverages in the portfolio at hand, we set the limit at $\omega_1 = \omega_2 = 730$ days, equivalent to two calendar years. Almost 94% of all Accident Benefits claims trigger that coverage within six months of reporting, and most of the remaining claims will be labelled as Accident Benefits no later than a year after the reporting date. The activation delays are longer for Bodily Injury claims, with 1.69% of them taking two years or more after reporting to be labelled as such. In our portfolio, a claim always triggers at least one coverage. As such, 22.4% of observed claims trigger both the Accident Benefits and Bodily Injury coverages, 55.4% of the claims only activate the Accident Benefits coverage and the remaining 22.2% of claims are solely Bodily

Injury claims. In other words, the levels of censoring observed for both coverages are of 22.2% and 55.4% for, respectively, Accident Benefits and Bodily Injury.

Table 7. Percentage of claims with different activation delays, shown in periods of six months.

Coverage	Activation delays				
	No delay	1 period	2 periods	3 periods	≥ 4 periods
Accident Benefits	93.84	5.73	0.29	0.08	0.06
Bodily Injury	85.86	9.86	1.46	1.13	1.69

6.3 Analysis

After using the approach described in Section 3 to find the non-parametric estimator of $\hat{K}_n(v)$ and $\hat{\psi}_n(v)$, we first find the parametric copula best fitted to our portfolio using the graphical procedure from Section 4. We present the plots of the non-parametric estimators $\hat{K}_n(v)$ and $\hat{\lambda}_n(v)$ in Figure 9, along with the corresponding fitted curves for four candidate Archimedean models. In both plots, the non-parametric estimator is closest to the curves of the Joe copula. With an estimated $\hat{\tau} = 0.2705$, the Joe copula with $\hat{\alpha}_j = 1.6652$ best fits our data.

The omnibus procedure first confirms this result. We estimate the dependence parameters for the competing models using the likelihood in Equation (9). The results shown in the third and fourth columns of Table 8 indicate that the Joe copula is most appropriate since it presents the smallest difference between $\hat{\alpha}$ and $\hat{\alpha}^*$. Similarly, the results of 1000 bootstrapped simulations for the L^2 -norm validation approach shown in the fifth column of Table 8 show that $D(\hat{\alpha})$ is the smallest for the Joe copula. We finally apply Wang 2010's goodness-of-fit test and find again in the last column of Table 8 that the Joe model presents the smallest percentage of rejection when we perform 1000 simulations for each competing model.

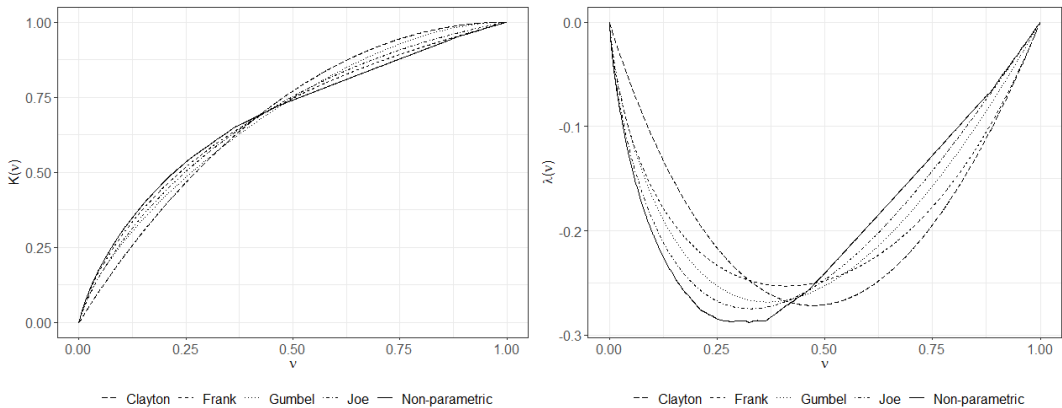


Figure 9. $K(v)$ and $\lambda(v)$ for the different copulas and for the Canadian automobile insurance dataset with a limit at 730 days.

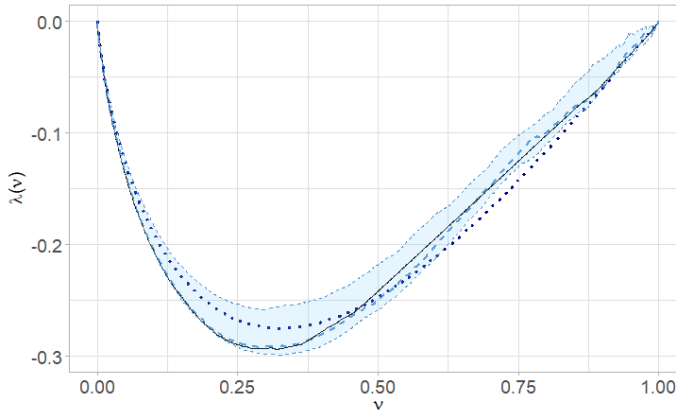
Even though the results from Table 8 indicate that the Joe copula is an appropriate fit for the data, we observe on the plot of the $\lambda(\cdot)$ functions in Figure 9 that the non-parametric $\hat{\lambda}_n(v)$, although being indeed closest to the curve $\lambda_{\hat{\alpha}_j}(v)$ for the Joe model, is still a little distance away from it. In this case, it might then be more fitting to directly work with the non-parametric generator $\hat{\psi}_n(v)$, using

Table 8. Results validation, based on 1000 bootstrapped simulations for the L^2 -norm and 1000 simulations for Wang 2010's test.

Copula	$\hat{\tau}$	Omnibus procedure		L^2 -norm $D(\hat{\alpha})$	Wang 2010 % rejection H_0
		$\hat{\alpha}^*$	$\hat{\alpha}$		
Clayton	0.2705	0.2432	0.7417	0.00399	0.9994
Frank		1.1166	2.5612	0.00180	0.9265
Gumbel		1.0554	1.3708	0.00084	0.8742
Joe		1.3821	1.6652	0.00033	0.0476

the algorithms described in Section 5. In what follows, we compare simulation results obtained with the Joe copula selected with the graphical procedure and by working directly from the generator function for the data at hand.

In Figure 10, we present a comparison between $\hat{\lambda}_n(v)$ (black continuous curve), the non-parametric estimator of $\lambda(\cdot)$ obtained for our insurance portfolio using the methodology laid out in Section 3, $\lambda_{\hat{\alpha}_J}(v)$ (dark blue dotted curve), the corresponding curve for the selected parametric Joe model, and $\lambda_{\hat{\psi}_n}(v)$ (light blue dashed curve), the curve obtained when re-simulating the data using the non-parametric generator $\hat{\psi}_n(v)$ and Algorithms 1 and 2. The dashed blue curve displayed on the plot is the average of performing 1000 simulations and the shaded area shows the distance between the smallest and highest curves among these simulations. We observe that the curve for the data simulated from the non-parametric generator is much closer to the curve for the original data than the one from the parametric Joe copula model. This indicates that the new simulated observations for the Accident Benefits and Bodily Injury activation delays have a dependence structure and a strength of dependence that are closest to the ones observed in the original data than the observations simulated using the parametric copula model.

**Figure 10.** Comparison between the non-parametric $\hat{\lambda}_n(v)$ estimated for the original data (blue continuous curve), the parametric $\lambda_{\hat{\alpha}_J}(v)$ (dark blue dotted curve) for the selected Joe copula model and $\lambda_{\hat{\psi}_n}(v)$ (light blue dashed curve) for the data simulated 1000 times using the generator $\hat{\psi}_n(v)$.

In Figures 11 and 12, we compare the results obtained from simulations using the parametric and non-parametric models. Figure 11 shows the heat maps for the scatterplots of the Accident Benefits and Bodily Injury activation delays, simulated from the parametric Joe copula model (left) and using the non-parametric estimator of the generator function (right). We observe that, although both heat maps display a high concentration of points in the top right corner of the plots, as is typically

observed with Joe copulas, this concentration seems higher for the non-parametric model. This would indicate a closer fit with the typical Joe copula scatterplot than for the data simulated from the parametric model. This is aligned with the results observed in Figures 9 and 10. In Figure 9, the curve of the Joe copula model is quite close to the curve for the Gumbel model. This could explain that we also observe a small concentration of points at the bottom left corner of the scatterplot for the parametric model in Figure 11: although we are indeed simulating from a Joe copula, the plots from Figure 9 indicate that this model is not too far from a Gumbel copula. In Figure 10, the curve for the non-parametrically simulated data is closer to the curve of the original data and more pulled towards the bottom left corner of the plot than the curve for the Joe copula. As such, the dependence structure usually observed in a Joe model seems more pronounced in the original data and in the non-parametric simulations than it is in the parametric simulations.

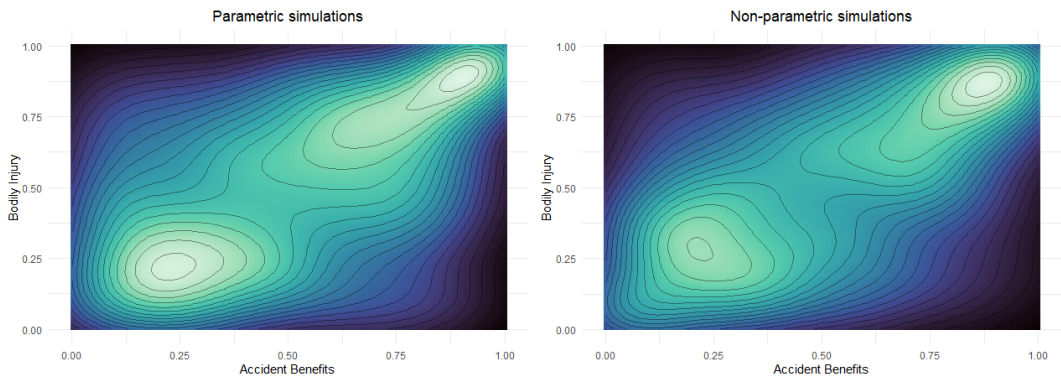


Figure 11. Heatmaps of the scatterplots for some simulations of the Accident Benefits and Bodily Injury activation delays, using the parametric Joe copula model (left) and the non-parametric generator function (right).

In Figure 12, we compare the original density functions of the Accident Benefits and Bodily Injury activation delays to the densities simulated using the parametric and non-parametric models. For both coverages, we observe that the densities for the non-parametric simulations (light blue dashed curves) are closest to the true densities (black continuous curves) than the densities for the parametric model (dark blue dotted curves). This is particularly true for the Bodily Injury coverage, where we see some very similar shapes between the black and light blue curves. The vertical lines represent the average observed activation delays (black), average simulated delays with the non-parametric model (light blue) and with the Joe copula (dark blue). These values are also given in Table 9. The Joe copula model appears to underestimate the activation delays for both coverages, particularly for Bodily Injury claims with a difference of over 60 days between an average simulated delay of 167.17 days compared to an observed average of 233.50 days. The difference is of almost 40 days for the Accident Benefits coverage, with an average simulated delay of 67.67 days compared to an observed delay of 107.12 days. The non-parametric model simulations provide better accuracy, with a close average simulated delay of 100.29 days for the Accident Benefits coverage and a slightly higher estimated delay of 245.12 days for Bodily Injury claims.

Table 9. Average activation delays (observed and simulated).

Coverage	Observed	$\hat{\psi}^{-1}(\nu)$	Joe copula
Accident Benefits	107.12	100.29	67.67
Bodily Injury	233.50	245.12	167.17

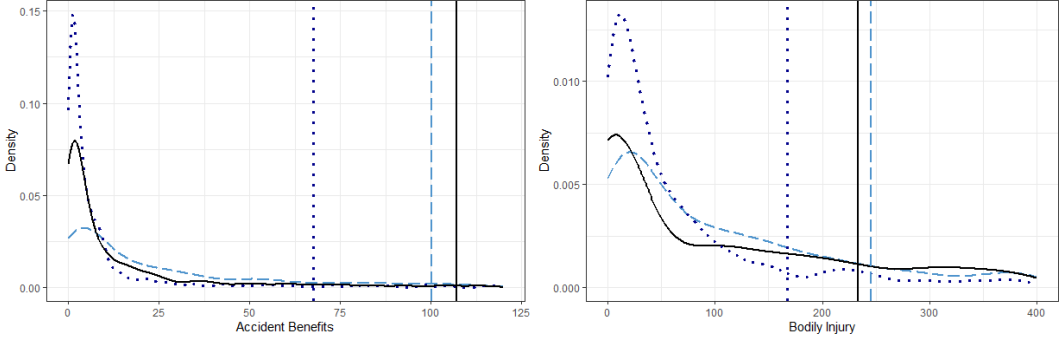


Figure 12. $K(v)$ and $\lambda(v)$ for the different copulas and for the Canadian automobile insurance dataset with a limit at 730 days.

6.4 A simple claims reserving example

To illustrate the usefulness of our model and the way it can be used as part of a larger framework, we provide in this section a simple reserves simulation scheme.

Assuming that a claim costs a fixed amount of 1\$, we use the classic exponential growth model to evaluate the portfolio reserve for different levels of inflation, taking January 1st, 2019 as valuation date. Let r denote the annual inflation rate. The annualised cost for claim i related to insurance coverage j , denoted by $Z_{i,j}$, is given by

$$Z_{i,j} = \left(1 + \frac{r}{360}\right)^{Y_{i,j}},$$

where $Y_{i,j}$ is the activation delay of coverage j for claim i .

Table 10 presents the results of 5 000 simulations for the Accident Benefits and Bodily Injury reserves, estimated using the parametric Joe copula models and the non-parametric model. We let the inflation r vary from 2% to 8% by increments of 2%. In each scenario, we observe that the simulated reserves with the non-parametric model are closer to the true amounts than those obtained with the parametric copula. This was to be expected, considering the results observed in Figures 10, 11 and 12. For the Accident Benefits coverages, the simulated reserves with the non-parametric model are slightly smaller than the true reserves but higher than the predictions obtained with the Joe copula. Note, however, than in a more realistic reserving setting where we would fit models for the claim severities, the small underestimation that we observe here would probably disappear or become insignificant. For the Bodily Injury coverage, non-parametric predictions are slightly larger than the true amounts while the predictions made from the Joe model underestimate the true reserves. These results are in line with what we observe in Table 9, where the predictions for the Accident Benefits delays are a bit smaller than the observed delay with the non-parametric model and even smaller with the parametric model. For the Bodily Injury coverage, the predictions for the delays are a bit higher for the non-parametric model and much lower for the Joe copula model.

Although we use here a very simplified model for illustration purposes, the results already hint that using the more accurate delay predictions from the non-parametric models will certainly lead to greater accuracy in the reserves predictions as well.

7. Conclusion

In this paper, we present a non-parametric estimator for the generator function of the Archimedean family of copulas, applicable to flexible censoring scenarios. Following the initial idea of Genest and

Table 10. Comparison between the true reserves and predicted reserves from the non-parametric and parametric models under different inflation scenarios.

Coverage	Model	Value	Inflation rate			
			0.02	0.04	0.06	0.08
Accident Benefits	Non-parametric	Observed	7593.67	7644.94	7697.88	7752.53
		Average	7585.83	7628.75	7672.76	7717.97
		VaR _{0.95}	7587.33	7631.95	7677.47	7724.44
	Parametric	Average	7578.17	7613.32	7649.56	7687.01
		VaR _{0.95}	7579.70	7616.51	7654.29	7693.47
	Observed		7640.68	7740.23	7842.75	7948.33
Bodily Injury	Non-parametric	Average	7648.70	7756.59	7867.75	7982.11
		VaR _{0.95}	7650.93	7761.01	7874.65	7991.37
	Parametric	Average	7618.66	7695.55	7774.71	7856.43
		VaR _{0.95}	7620.71	7699.67	7780.94	7865.09
	Observed		7640.68	7740.23	7842.75	7948.33
	Average		7648.70	7756.59	7867.75	7982.11

Rivest 1993 and combining it with tools from survival analysis in order to account for the presence of incomplete data, we derive the estimator $\hat{\psi}_n(\gamma)$.

We then present two approaches to use it. We begin with a graphical procedure that compares the estimator for a given dataset to different parametric copula models, and select the best fitted one among them. Thanks to our extension of the non-parametric estimator to various censoring schemes, we show that this simple graphical selection method works well with both complete and incomplete data by illustrating three results validation approaches.

Then, rather than seeking to use a well-defined parametric copula model, we suggest to directly work from the non-parametric estimator of the generator function to perform simulations and further analysis on the data. This approach is interesting, for example, when the graphical comparison does not lead to a clear resemblance between the non-parametric curve and one of the candidate copula models. It allows to forego any parametric assumptions and work instead directly from the non-parametric estimator without labelling it to any Archimedean family. Viewing the generator as the Laplace-Stieltjes transform of a strictly positive random variable, we propose to use the RLAPTRANS algorithm of Ridout 2009 in combination with an Archimedean copula sampling algorithm like the one suggested by Marshall and Olkin 1988. With just a small modification of the RLAPTRANS algorithm to use the non-parametric function rather than a parametric one, we are able to generate random observations using the inverse Laplace-Stieltjes transform. We then use these random observations to generate couples of observations with the same shape and strength of dependence as those in the original data. Considering that the non-parametric estimator we derive for the generator function takes into account the flexible censoring scenarios that can be found in the data, this method can also be used in the presence of incomplete data. We demonstrate this by means of some simulations studies.

We then apply our methods to an automobile claims insurance portfolio for which we seek to model the dependence between two insurance coverages by means of their activation delays. These correspond to the time elapsed between the reporting date of a claim and the date at which the claim triggers one of the coverages under which the policyholder is insured. Our data presents two levels of censoring. First, if a coverage does not become active before the claim settles, then the activation delay for this specific coverage is censored. Second, considering current insurance legislation in Canada and the specific nature of our claims, we apply a limit of 730 days after which, even if a claim is still open, additional coverages can not be triggered anymore. This limit implies that after a certain amount of time has passed, policyholders are not able to seek new compensation anymore, even if their claim is still open. This acts as a second level of censoring on the activation delays of the coverages. We find non-parametrically the generator function in this specific censoring scenario

and then compare the estimation results when using both the best-fitted parametric copula model, selected via the graphical procedure, and direct simulations from the generator. In this specific case, even though the Joe copula with parameter $\hat{\alpha} = 1.6652$ is deemed well-fitted to the data by the three model validation approaches, there exists a small gap on the plot of the $\lambda(\cdot)$ functions in Figure 9 between this parametric model and the curve for the data. We show that in this case, directly simulating from the generator function using Algorithms 1 and 2, leads to more accurate predictions of the activation delays than when using the Joe copula model. We conclude by a simple illustration of how these results can then be used as part of a larger claims reserving framework. We demonstrate that using the predictions from the non-parametric model can lead to reserves predictions that are closer to the true reserves amounts.

This approach could be used as part of a larger model such as the one proposed in Michaelides, Pigeon, and Cossette 2023 or Côté, Genest, and Stephens 2022 where the estimation of the activation delays can replace, respectively, the activation patterns and the multinomial estimation of the claim type. A similar methodology could also be used in different models such as that proposed in Antonio and Plat 2014 where, for example, the next event's exact time for reported but not settled claims could be extended to a multivariate framework by considering simultaneously, as in Section 6, different business lines or insurance coverages.

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Appendix 1. Useful functions and relationships for some popular Archimedean copulas

Table 11. Expressions of the copula, generator function and relation between α , Kendall's tau and $\lambda(\cdot)$ for some commonly used Archimedean copulas.

Copula	$C_\alpha(u_1, u_2)$	$\psi(v)$	τ	$\lambda_\alpha(v)$
Clayton	$(u_1^{-\alpha} + u_2^{-\alpha} - 1)^{-1/\alpha}$	$(1 + t)^{-1/\alpha}$	$\alpha/(\alpha + 2)$	$v(v^\alpha - 1)/\alpha$
Frank	$-\frac{1}{\alpha} \ln \left(1 + \frac{(e^{-\alpha u_1} - 1)(e^{-\alpha u_2} - 1)}{e^{-\alpha} - 1} \right)$	$-(\ln(e^{-t} - 1) + 1)/\alpha$	$1 + \frac{4}{\alpha} \left(\int_0^\alpha \frac{\xi}{\alpha(e^\xi - 1)} d\xi - 1 \right)$	$-\ln \left\{ (e^{-\alpha v} - 1)/(e^{-\alpha} - 1) \right\} / (-\alpha/(e^{\alpha v} - 1))$
Gumbel-Hougaard	$\exp(-[(\tilde{u}_1)^\alpha + (\tilde{u}_2)^\alpha]^{1/\alpha})$	$\exp(-t^{1/\alpha})$	$1 - 1/\alpha$	$v \ln(v)/\alpha$
Joe	$1 - (\tilde{u}_1^\alpha + \tilde{u}_2^\alpha - \tilde{u}_1^\alpha \tilde{u}_2^\alpha)^{1/\alpha}$	$1 - (1 - e^{-t})^{1/\alpha}$	$-$	$-\ln \left\{ 1 - (1 - v)^\alpha \right\} / - \left((\alpha(1 - v)^{(\alpha-1)}) / (1 - (1 - v)^\alpha) \right)$

Appendix 2. Model validation approaches for the parametric copula selected via the graphical procedure

Omnibus estimation procedure

The omnibus estimation procedure is a semi-parametric optimization procedure that substitutes empirical versions of the marginal distributions in the (parametric) likelihood function of the copula model, $L(\cdot)$.

We use rescaled Kaplan–Meier estimators for T_1 and T_2 , multiplying the original Kaplan–Meier versions by $n/(n+1)$, where n is the total number of observations in the sample. As mentioned in Genest, Ghoudi, and Rivest 1995 as well as Denuit, Purcaru, and Van Keilegom 2006, this rescaling prevents issues related to the potential unboundedness of the copula log-density as u_1 and u_2 tend to one.

Once we have the marginal distribution functions, we use them in the estimation step, i.e., the optimization of the model likelihood to obtain the pseudo-likelihood estimator $\hat{\alpha}^*$ of the dependence parameter:

$$\hat{\alpha}^* = \arg \max L(u_1, u_2, \delta_1, \delta_2; \alpha),$$

where the likelihood function under flexible censoring scenarios is given by:

$$\begin{aligned} L(u_1, u_2; \delta_1, \delta_2; \alpha) = & \prod_{i=1}^n c_{\alpha}(u_{1i}, u_{2i}; \alpha)^{\delta_{1i}\delta_{2i}} + \left(\frac{\partial C_{\alpha}(u_{1i}, u_{2i}; \alpha)}{\partial u_1} \right)^{\delta_{1i}(1-\delta_{2i})} \\ & + \left(\frac{\partial C_{\alpha}(u_{1i}, u_{2i}; \alpha)}{\partial u_2} \right)^{(1-\delta_{1i})\delta_{2i}} + C_{\alpha}(u_{1i}, u_{2i}; \alpha)^{(1-\delta_{1i})(1-\delta_{2i})}. \end{aligned} \quad (9)$$

C_{α} is the candidate Archimedean copula under consideration, c_{α} its density and $\delta_{ji} = 1$ if $T_j < X_j$ for $j = 1, 2$ for observation i and equals zero otherwise (i.e., for a censored observation). The partial derivatives with respect to either u_1 or u_2 can be found in Table 12.

Table 12. Partial derivatives for some popular Archimedean copulas with $\tilde{u} = -\ln u$ and $\bar{u} = 1 - u$.

Copula	$\partial C_{\alpha}(u_1, u_2)/\partial u_1$
Clayton	$[1 + u_1^{\alpha}(u_2^{-\alpha} - 1)]^{-1-1/\alpha}$
Frank	$[e^{-\alpha u_1} - e^{-\alpha(u_1+u_2)}] \times [(1 - e^{-\alpha}) - (1 - e^{-\alpha u_2})(1 - e^{-\alpha u_1})]^{-1}$
Gumbel	$u_1^{-1} e^{-(\tilde{u}_2^{\alpha} + \tilde{u}_1^{\alpha})/\alpha} [1 + (\tilde{u}_2/\tilde{u}_1)^{\alpha}]^{-1+1/\alpha}$
Joe	$(1 - \bar{u}_2^{\alpha})(1 - \bar{u}_2^{\alpha} + \bar{u}_2^{\alpha}\bar{u}_1^{-\alpha})^{-1+1/\alpha}$

In order to use the omnibus procedure to validate the model choice resulting from the graphical comparison, we retrieve the pseudo-maximum likelihood estimator $\hat{\alpha}^*$ for the chosen candidate copula models and compare them to the estimators $\hat{\alpha}$ found using Equation (7). The copula with the smallest difference between these two estimators is the most appropriate for the data.

L^2 -norm

The second approach to validate the choice of copula model is based on the L^2 -norm. We follow the methodology laid out in Wang and Wells 2000 that introduce a goodness-of-fit statistic specifically designed for Archimedean copulas with censored data under the assumption that a consistent non-parametric estimator of the bivariate joint distribution function is available. Note that with estimated parameters present in the specification of the null hypothesis for this test, the approach we show

here is not an actual statistical test but rather a way to validate our model. However, we will use the word *test* in this section for simplicity. As mentioned in the introduction, more formal and general methods have been proposed in the literature. We discuss one in the next paragraph.

Wang and Wells 2000's test relies on the L^2 -norm distance between the empirical estimator $\hat{K}_n(v)$ and the corresponding $K_{\hat{\alpha}}(v)$ of the hypothesized parametric copula model,

$$D(\hat{\alpha}) = \int_{\xi}^1 (\hat{K}_n(v) - K_{\hat{\alpha}}(v))^2 dv.$$

We use Riemann sum approximations to simplify numerical analysis:

$$D(\hat{\alpha}) = \sum_{i=1}^n (\hat{K}_n(v_{(i)}) - K_{\hat{\alpha}}(v_{(i)}))^2 (v_{(i)} - v_{(i-1)}),$$

where $v_{(i)}$ is the i^{th} ordered value of $\{v_1, v_2, \dots, v_n\}$. However, the asymptotic distribution of $D(\hat{\alpha})$ is difficult to retrieve analytically.

Goodness-of-fit test for censored dependent data

The omnibus procedure and L^2 -norm validation approach are easy to implement and give a good idea of whether a copula model is appropriate for the data at hand. However, neither are formal statistical tests. To further increase the confidence in our choice of copula model, we present a more formal approach for Archimedean copulas, proposed by Wang 2010.

This goodness-of-fit test applies to uncensored and right-censored data thanks to a multiple imputation procedure. It bases its premise on Proposition 1.1. from Genest and Rivest 1993.

For a bivariate vector of dependent observations (T_1, T_2) whose marginal survival functions $S_1(\cdot)$ and $S_2(\cdot)$ have a dependence function $C(\cdot)$ of the form

$$C(S_1(t_1), S_2(t_2)) = \psi \left\{ \psi^{-1}(S_1(t_1)) + \psi^{-1}(S_2(t_2)) \right\},$$

then, the random variables

$$U = \frac{\psi^{-1}(S_1(t_1))}{\psi^{-1}(S_1(t_1)) + \psi^{-1}(S_2(t_2))} \quad \text{and} \quad V = \psi \left\{ \psi^{-1}(S_1(t_1)) + \psi^{-1}(S_2(t_2)) \right\} = C(S_1(t_1), S_2(t_2)). \quad (10)$$

are independent. In other words, under the null hypothesis that an Archimedean copula can model the dependence between (T_1, T_2) with generator $\psi(\cdot)$, the correlation coefficient between U and V is null:

$$H_0 : \rho = 0 \quad \text{vs} \quad H_1 : \rho \neq 0.$$

Let

$$r_n = \frac{\sum_{i=1}^n (\hat{U}_i - \bar{\hat{U}})(\hat{V}_i - \bar{\hat{V}})}{\sqrt{\sum_{i=1}^n (\hat{U}_i - \bar{\hat{U}})^2 \sum_{i=1}^n (\hat{V}_i - \bar{\hat{V}})^2}},$$

where $\hat{U}$ and $\hat{V}$ are consistent estimators of U and V and where $\bar{\hat{U}}$ and $\bar{\hat{V}}$ are the sample means of $\hat{U}_i$ and $\hat{V}_i$. The test statistic is defined as

$$Z_n = \frac{1}{2} \log \left[\frac{1 + r_n}{1 - r_n} \right]$$

and we have that $\sqrt{n}Z_n \rightarrow N(0, 1)$ in distribution.

For censored data, we can not directly retrieve $\hat{U}_i$ and $\hat{V}_i$ from the estimators proposed by Genest and Rivest 1993 based on Equation (10). Instead, Wang 2010 shows that, depending on the censoring pattern, the data can be simulated using a multiple imputation procedure from one of the distributions presented below. All proofs and further details are in Wang 2010.

Let (T_1, T_2) be a bivariate vector submitted to censoring by the bivariate continuous vector (X_1, X_2) , as described in Section 3.1. Let $S_1(t_1)$ and $S_2(t_2)$ be the marginal survival functions of, respectively, T_1 and T_2 , whose dependence function can be modelled by an absolutely continuous Archimedean copula such that

$$C(S_1(t_1), S_2(t_2)) = S(t_1, t_2) = \psi \left\{ \psi^{-1}(S_1(t_1)) + \psi^{-1}(S_2(t_2)) \right\}.$$

Then, we have that:

1. if both observations are censored, the distribution function of $(V|T_1 > x_1, T_2 > x_2)$ is

$$F_1(v, x_1, x_2) = \frac{1}{S(x_1, x_2)} \left[v - \frac{\psi^{-1}(v) - \psi^{-1}\{S(x_1, x_2)\}}{\psi^{-1'}(v)} \right], \quad 0 \leq v \leq S(x_1, x_2)$$

and the distribution function of $(U|T_1 > x_1, T_2 > x_2)$ is uniformly distributed on the interval

$$\left[\frac{\psi^{-1}\{S_1(x_1)\}}{\psi^{-1}(v)}, 1 - \frac{\psi^{-1}\{S_2(x_2)\}}{\psi^{-1}(v)} \right];$$

2. if only the second observation is censored, the distribution function of $(V|T_1 = t_1, T_2 > x_2)$ is

$$F_2(v, t_1, c_2) = \frac{p'(\psi^{-1}(v))}{p'(\psi^{-1}(S(t_1, x_2)))}, \quad 0 \leq v \leq S(t_1, x_2)$$

with $p(\cdot) = \psi(\cdot)$ and the distribution function of $(U|T_1 = t_1, T_2 > x_2)$ is

$$G_2(y, t_1, x_2) = \frac{p'\{\psi^{-1}(S_1(t_1))/u\}}{p'\{\psi^{-1}(S(t_1, x_2))\}}, \quad 0 \leq u \leq \frac{\psi^{-1}\{S_1(t_1)\}}{\psi^{-1}\{S(t_1, x_2)\}};$$

3. if only the first observation is censored, the distribution function of $(V|T_1 > x_1, T_2 = x_2)$ is

$$F_3(v, x_1, t_2) = \frac{p'(\psi^{-1}(v))}{p'(\psi^{-1}(S(x_1, t_2)))}, \quad 0 \leq v \leq S(x_1, t_2)$$

and the distribution function of $(U|T_1 > x_1, T_2 = t_2)$ is

$$G_3(y, x_1, t_2) = \frac{p'\{\psi^{-1}(S_2(t_2))/(1-u)\}}{p'\{\psi^{-1}(S(x_1, t_2))\}}, \quad \frac{\psi^{-1}\{S_1(x_1)\}}{\psi^{-1}\{S(x_1, t_2)\}} \leq u \leq 1;$$

4. if both observations are uncensored, the distribution function of $(V|T_1 = t_1, T_2 = t_2)$ is the same as in Proposition 1.1. from Genest and Rivest 1993 and the conditional distribution of $(U|V = v)$ is just a uniform distribution on $[0, 1]$ independent of V .

Appendix 3. Simulation study: Results of the Omnibus procedure

For each of the three scenarios, Table 13 displays the results of the omnibus procedure. For each simulated sample, we compare the dependence parameters $\hat{\alpha}$ retrieved from Equation (7) to the maximum likelihood estimator $\hat{\alpha}^*$. We observe that in most cases, the distance between $\hat{\alpha}$ and $\hat{\alpha}^*$ is minimal for the correct copula, i.e., the omnibus procedure leads to selecting the copula model from which the data was originally simulated. We count three occurrences where the procedure selects an alternative model. In the single-censoring scenario, the Gumbel model is chosen for the Clayton sample, and the Joe copula is selected for the Gumbel sample. The Gumbel copula is selected for the Frank sample in the double-censoring scenario. However, we observe that the correct copulas present in each case are the second smallest distance between $\hat{\alpha}$ and $\hat{\alpha}^*$ and that the difference with the third and fourth distances is quite marked. For the Gumbel sample in the single censoring scenario for example, the smallest distance between the parameters is equal to 0.0428 for the Joe copula, 0.0992 for the Gumbel copula and then, 0.3988 and 0.4971 for, respectively, the Clayton and Frank copulas.

Table 13. Omnibus procedure for different censoring scenarios.

True copula	Candidate model	No censoring		single-censoring		double-censoring	
		$\hat{\alpha}$	$\hat{\alpha}^*$	$\hat{\alpha}$	$\hat{\alpha}^*$	$\hat{\alpha}$	$\hat{\alpha}^*$
Clayton ($\alpha = 1.3332$)	Clayton	1.7694	1.8284	1.1789	1.3263	0.9240	0.8541
	Frank	5.2063	5.7575	3.7724	4.1154	2.1323	2.7655
	Gumbel	1.8846	1.7294	1.5895	1.4926	1.4621	1.3070
	Joe	2.6347	1.7373	2.0732	1.4713	1.5231	1.2792
Frank ($\alpha = 4.1611$)	Clayton	1.3202	0.8049	1.2055	0.7428	1.0034	0.5326
	Frank	4.1284	4.2527	3.8403	3.9827	3.3149	2.9126
	Gumbel	1.6602	1.5274	1.6029	1.4533	1.5018	1.1694
	Joe	2.2066	1.6870	2.0983	1.6260	1.9084	1.1759
Gumbel ($\alpha = 1.6667$)	Clayton	1.1377	0.6557	1.1380	0.7392	0.9464	0.6493
	Frank	4.1641	4.3241	3.6674	4.1645	2.8121	3.6805
	Gumbel	1.5690	1.5934	1.5691	1.6683	1.4734	1.5283
	Joe	2.0345	1.8735	2.0347	1.9919	1.8671	1.7554
Joe ($\alpha = 2.2191$)	Clayton	1.2609	0.6067	1.1494	0.5514	0.9344	0.5170
	Frank	3.9802	4.2203	3.6968	3.8131	3.1298	3.4508
	Gumbel	1.6305	1.7284	1.5748	1.6356	1.4674	1.5638
	Joe	2.151	2.1997	2.0454	2.0431	1.8441	1.9360